

Q1. How many Pillars of OOPs? Name them.

=> OOps have Four Pillars-

1. Inheritance: ✓

- a. simple or single inheritance ✓
- b. multiple inheritance ✓
- c. multilevel inheritance ✓
- d. hybrid inheritance ✓
- e. hierarical inheritance

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2. Polymorphyism

- a. overriding ✓
- b. pverloading ✓

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3. Encapsulation

- a. Public ✓
- b. Protected ✓
- c. Private ✓

4. Data Abstraction ✓

Q2. What is Overriding and Overloading.

=> 1. Overloading - In Overloading it has the same method in the single or same class with different parameters. ✓

2. Overriding- In Overriding it has the same method name of parent class with respect to child class and provide its own implementation. ✓

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Q3. what is Encapsulation in OOPs?

=> Encapsulation in oops is wrapping or securing the data/method or restrict from accessing by third person.

it has three types:

- a. Public
- b. Protected
- c. Privat

✓

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Q4. How many types of Inheritance? Name them.

=> Inheritance has Five types :

- a. simple or single inheritance
- b. multiple inheritance
- c. multilevel inheritance
- d. hybrid inheritance
- e. hierarical inheritance



Q5. How many types of Case? Name them and give one example of each?

=> There are three types of cases in python:

- a. Pascal case - All words First letter is capital
ex-StudentDetails



- b. camel case - first word's letter is small and rest word's letters are capital
ex-addvalue



- c. snake case - word's letter are in small case but has separated by underscore
ex-length_breadth



Q6. What is class in OOPs and write some key points.

=> -Class in OOPs is also known as Blueprint:

- It is used as key name (class) to access the OOPs functionality
- it includes methods in it



Q7. What is object in OOPs and some key points?

=> -object in OOPs is known as thing's

- it is instance of any class
- it occupies the storage to store the data
- it can access the class variable and methods



Q8. How many types of Variables in OOPs?

=> There are two types of variables in OOps:

1. Instance Variable

- it can access through self keyword but need `__init__` constructor to create
- gets the data of the same object
`self.name=name`



2. Class variable

- it can access through cls keyword
- data accessible to different object
- data is in the class but same for different objects
`cls.college=college name`



Q9. How many types of Methods in OOPs Name Them?

=> There are three types of method in OOPs :

1. Instance method

- it can access through self keyword and it has no decorator

2. Class method

- it can access through cls keyword and decorator @classmethod

3. static method

- doesn't have any keyword just use decorator @staticmethod

(self)
✓
(cls)
✓
(2)
✓

Q11. Write Rules of Overriding?

=> In Overriding it has the same method name of parent class with respect to child class and provide its own implementation.

- method names should be same ✓

- parameters should be same ✓

- need to have different class

ex- class animal():

def sound(self):

print("animal sound")

class dog(animal):

def sound(self):

print("dog barks")

class cat(animal):

def sound(self):

print("cat meow")

d=dog()

d.sound

c=cat()

c.sound

(6)
✓

Q12. What is super () in OOPs

=> super() is a built in function in python to access the method and variables from parents class to child class.

ex- class employee():

def work(self):

print("need to do hardwork")

class manager(employee):

def job(self):

print("Very responsible job")

super().work()

✓

✓

(2)
✓

m = Manager()

m.job()

Q13. How many steps to read a file using python? Write steps?

=> to read a file in python:

1. first create a variable store the file path in it ✓
2. then open the variable/file using open() with mode 'r' ✓
3. then read the file using read() ✓
4. print the content of that file ✓
5. close the file using close() ✓

2

with()

Q14. How many steps to write a file using python? Write steps?

=> to write a file in python:

1. first create a variable store the file path in it
2. then open the variable/file using open() with mode 'w'
3. then write the file using write()
4. close the file using close()

2

#Q15. Write code to checking files is exists or not in your pc drive?

file=r"C:\Users\Yash\OneDrive\Desktop\check1.txt"

try:

```
file1=open(file,"r")
print("file exist")
content=file1.read()
print(content)
file1.close()
```

except FileNotFoundError:

```
print("file does not exists")
```

if os.file.exists()

#Q21. Extract all vowels from the string using list comprehension: text

"Comprehension"?

name1="Comprehension"

vowels=[i for i in name1 if i in "aeiouAEIOU"]

print(vowels)

2